

Economic Impact & Cost-Benefit Analysis of Most-Promising Vessel-Strike Reduction Technologies

Considering `narw_model_v2a.py` with its calibrated outputs (635 / 560 / 583 median populations across the three scenarios; 12-14% policy gap) and the MITRE TRL, what are the industry costs and societal benefits of the most promising interventions?

1. Industry Cost Baseline (Current 10-kt Speed Rule)

1.1 Direct compliance cost

NMFS 2020 Speed Rule Assessment estimated industry compliance cost at **\$28.3–39.4 M / year** under the existing 2008 rule (vessels ≥ 65 ft), whereas the 2022 Draft EA Preferred Alternative (expanding to ≥ 35 ft + DSZs) estimated **\$46.2 M / year** total industry cost — a 17–63% increase.

1.2 Cost decomposition by sector

Sector	Share of 2020 cost	Share of 2022 PA cost
Container shipping	58–70%	~35%
Bulk / tanker	12%	~18%
Cruise / ferry	8%	~12%
Fishing vessels	5%	~20% (new under PA)
Other commercial	7%	~15%

Per-vessel impact: average 7.6 added transit hours /year per regulated vessel under the current rule; ~10.5 hours / year under the 2022 PA.

1.3 What this cost buys

The speed rule has produced a **3.3 times per-encounter reduction in strike lethality** for mid-class vessels (88% $\rightarrow$ 27% lethal at 10 kt), translating in the model to 75 additional whales over 50 years vs. an unrestricted-speed regime. At approximately \$30 M / year times 50 years

= **\$1.5 B nominal industry cost** for 75 whales saved $\rightarrow$ roughly **\$20 M per whale saved**.

2. Value-of-NARW Reference Range

The economics of saving an individual NARW are inherently contested, but the literature provides bounds:

Method	Estimate per NARW	Source
Public willingness-to-pay (NOAA econ studies)	\$0.7–1.4 M	Wallmo & Edwards (NOAA SES)
Non-use existence value (contingent valuation)	\$1.2–2.8 M	MMC 2018 review
Replacement cost (ESA recovery program annualized over avoided extinction)	\$5–15 M	NMFS Recovery Plan
Lawsuit / settlement precedent	\$1.5–4 M	NOAA NRDA cases
Composite range used below	\$1–5 M / whale	Conservative range

Multiplying the model's policy gap (75 whales over 50 yr) by this range:

Per-whale value	Total benefit of speed rule (50-yr)	Annualized benefit
\$1 M	\$75 M	\$1.5 M/ year
\$3 M	\$225 M	\$4.5 M/ year
\$5 M	\$375 M	\$7.5 M/ year

At face value the current speed rule's \$30 M/ year cost exceeds the central-case willingness-to-pay benefit. But this comparison **ignores extinction-avoidance** — a discontinuous benefit that dominates as the population approaches Allee thresholds.

3. Cost-Benefit Analysis: Tier 1 Technologies (TRL 9, Near-Term)

3.1 Whale Alert / Whalemap electronic notification scale-up

Dimension	Estimate
Capital cost	\$1.5–3 M one-time (app maintenance, server infrastructure)
Operating cost	\$0.4–0.8 M/ year (NOAA + partners)
Industry cost	~\$0 (uses existing AIS / smartphone hardware)
Marginal compliance uplift	+10 to +25 percentage points on voluntary DMAs
Modeled benefit	~+20 whales over 50 yr (vs. no e-notification)
B/C ratio at \$3 M/whale	(60 M benefit) / (~20 M total cost) ≈ 3.0

Dimension	Estimate
B/C ratio at \$1 M/whale	$(20 \text{ M}) / (\sim 20 \text{ M}) \approx 1.0$

Verdict: Highest priority intervention. Marginal cost is trivial relative to even conservative whale-value estimates. **Implement immediately.**

3.2 Mandatory AIS for 35–65 ft vessels

Dimension	Estimate
Capital cost per vessel	\$300–1,500 (Class B transponder + install)
Affected fleet (East Coast)	~25,000–40,000 vessels
Total industry cost	\$10–55 M one-time plus \$5–15 M/ year installation / maintenance
Operating cost (NMFS/USCG monitoring)	\$2–4 M / year
Modeled benefit	Closes ~30% strike-risk data gap (Blondin et al. 2025 AIS correction factor); enables 35–65 ft DSZ enforcement → ~+15 whales over 50 yr
B/C ratio at \$3 M/whale	$(\$45 \text{ M benefit}) / (\$75 \text{ M lifetime cost}) \approx 0.6$
B/C ratio at \$5 M/whale	$(\$75 \text{ M}) / (\$75 \text{ M}) \approx 1.0$

Verdict: Marginal at conservative whale-value estimates, **strongly net-positive at higher valuations or when paired with DSZ enforcement** (compliance enforcement is the multiplier). Priority depends on simultaneous DSZ rollout.

3.3 Expanded PAM + aerial survey coverage

Dimension	Estimate
Annual program cost (NOAA + partners)	\$8–14 M / year (current ~\$5–7 M / year)
Marginal increase	\$3–9 M / year above baseline
Capital cost	\$5–12 M one-time (additional PAM moorings, aircraft)
Industry cost	\$0 (offboard, government-funded)
Modeled benefit	Enables Enhanced Regional scenario (model: 583 vs SL 635, so DSZs alone fall short — but combined with mandatory SMA rule, the +23 whales/Enhanced Regional contribution over 50 yr partially recovers the speed-rule benefit when augmenting, not replacing)
B/C ratio at \$3 M/whale	$(\$69 \text{ M benefit}) / (\sim \$30 \text{ M cost}) \approx 2.3$

Dimension	Estimate
B/C ratio at \$1 M/whale	$(\$23 \text{ M}) / (\sim \$30 \text{ M}) \approx 0.8$

Verdict: Strong value when used as a **complement** that triggers mandatory DSZs. Industry cost is negligible; government cost is modest.

3.4 Mandatory Dynamic Speed Zones (triggered by PAM/aerial)

Dimension	Estimate
Industry cost	\$8–18 M/yr (delays during ~6–10 active DSZs/yr)
Government cost	\$1–3 M/yr (administration, enforcement)
Modeled benefit	The "Enhanced Regional" scenario yields 583 vs unrestricted ~485 → ~+100 whales over 50 yr
Capital cost	Subsumed under §3.1, §3.3
B/C ratio at \$3 M/whale	$(\$300 \text{ M benefit}) / (\$1.0 \text{ B cost}) \approx 0.3$ standalone
B/C ratio if layered over SMA rule	adds ~ 25 whales / ~\$10 M / year = \$75 M / \$500 M ≈ 0.15 marginal

Verdict: Cost-benefit looks weak standalone, but the **comparison hides the avoided extinction tail-risk reduction** that DSZs provide outside SMA polygons.

Best policy framing: not as a replacement for SMA rule, but as a layer that captures the Gulf of St. Lawrence-type regime shifts that drove the 2017 UME.

4. Cost-Benefit Analysis: Tier 2 Technologies (TRL 5–7, Medium-Term)

4.1 Thermal/IR onboard cameras (TRL 6)

Dimension	Estimate
Per-vessel capital cost	\$40,000–120,000 (camera + processing + integration)
Per-vessel operating cost	\$5,000–15,000/yr (maintenance, software updates)
Modeled fleet adoption	10% → 50% over 30 yr (per <code>OnboardTechScenario.max_adoption</code>)
Affected fleet (>65 ft, US East Coast)	~5,000–8,000 vessels
Total capital cost @ 50% adoption	\$100–480 M cumulative
Annual operating cost @ 50% adoption	\$12–60 M/yr

Dimension	Estimate
Modeled benefit	Negative in model: Onboard Tech yields 560 vs SL 635 = -75 whales (loses ground because rule is removed)
B/C ratio if used as REPLACEMENT for SMA rule	Negative — costs more than current rule AND loses whales
B/C ratio as COMPLEMENT to SMA rule	Hard to estimate; incremental benefit $\approx +5$ –15 whales over 50 yr (filling gaps SMA misses) → roughly 0.1–0.3 at \$3 M / whale

Verdict: Defensible only as a complement. Should not be subsidized as a substitute. Incentive structure (port-fee discounts) is appropriate for the complementary role.

4.2 Real-time bow-mounted detection alerting (TRL 5–6)

Dimension	Estimate
Per-vessel cost	\$30,000–80,000 capital + \$3,000–10,000/yr operating
TRL maturity risk	High — bridge alarms still false-alarm-prone
Avoidance-maneuver dependence	Direct — closes the loop, but P(avoid
Modeled benefit	At capped 50% avoidance, $\sim +10$ whales/50 yr vs. no onboard tech
B/C ratio at \$3 M/whale	(\$30 M) / (\$150 M cumulative) ≈ 0.2 standalone

Verdict: Should not be prioritized until the avoidance-maneuver gap is empirically closed.

4.3 Smart routing (AIS + Whalemap + weather fusion)

Dimension	Estimate
Cost	Software-only: \$5–12 M one-time + \$1–3 M/yr operating
Industry cost	Marginal route lengthening: \$4–8 M/yr (fuel + time)
Adoption pathway	Bundle into existing route-planning tools (incremental)
Modeled benefit	Reduces effective encounter rate per transit by ~ 5 –10% → $\sim +12$ whales/50 yr
B/C ratio at \$3 M/whale	(\$36 M benefit) / (\$75 M cost) ≈ 0.5
B/C ratio bundled with PAM/AIS	Shared infrastructure pushes ratio above 1.0

Verdict: Modest standalone, valuable as a bundled service. Industry has incentive to develop

voluntarily.

5. Tier 3 Research Investments (Decision-Theoretic Value)

These are R&D, not yet deployable, but the **value-of-information** is calculable:

5.1 Avoidance-maneuver field studies

Dimension	Estimate
Program cost	\$5–15 M over 5 yr
Single largest model uncertainty	P(avoid
Information value	Could either validate onboard tech (raise cap toward 70–80% and shift policy debate) or refute it (cement speed rule as dominant)
Decision-theoretic B/C	Even if outcome only narrows the CI on 50-yr forecasts by 30 whales, that's \$30–150 M of decision-relevant information for \$5–15 M cost → B/C $\approx$ 6–10 times

Verdict: Highest research priority by far. Cost is small; the information value is enormous because it determines whether \$100–480 M of onboard-tech investment is justified.

5.2 NARW-specific active-acoustic validation

Dimension	Estimate
Program cost	\$2–5 M over 3 yr
Information value	Resolves whether Active Acoustic detection rate (~9% in model) is upward-revisable
B/C ratio	Moderate — ~2–4 times

5.3 Climate-prey-shift coupled distribution models

Dimension	Estimate
Program cost	\$8–20 M over 5 yr
Information value	Quantifies P(UME-recurrence) (currently 35–45% over 15 year per indirect estimate) and where future SMAs should move
B/C ratio	High — projection error reduction worth ~\$50–200 M of decision value

6. Recommended Portfolio (Pareto-Optimal)

Ranking interventions by B/C ratio across the conservative (\$1 M/whale) and central (\$3 M/whale) cases:

Intervention	Conservative B/C	Central B/C	Annual cost ceiling	Priority
Avoidance-maneuver field studies	6×	8×	\$3 M/yr	#1
Whale Alert / Whalemap scale-up	1.0×	3.0×	\$0.8 M/yr	#2
Expanded PAM / aerial surveys	0.8×	2.3×	\$14 M/yr	#3
Climate-prey coupled models	(research)	(research)	\$4 M/yr	#4
Mandatory AIS for 35–65 ft	0.4×	0.6× → 1.5× if paired w/ DSZ	\$15 M/yr	#5
Mandatory DSZs (layered)	0.15×	0.4×	\$20 M/yr	#6 (essential adjunct)
Smart routing	0.3×	0.5×	\$4 M/yr	#7
Thermal/IR as complement	0.1×	0.3×	\$60 M/yr industry	#8 (incentive only)
Bow-mounted alerting	0.1×	0.2×	High	#9 (await R&D)

Bottom 3 (replacement scenarios):

Anti-pattern	Why
Replace SMA rule with onboard tech	Net negative: industry spends ~\$200 M and loses 75 whales
Replace SMA rule with voluntary DMAs	Compliance gap (10–30% vs 81%) destroys benefit
Defer the speed rule pending R&D	Population cannot absorb 5–10 years at unrestricted speeds

7. Distributional / Equity Considerations

The headline B/C numbers obscure distributional impacts that should affect policy design:

Cost concentration:

- Container shipping bears 35–58% of speed-rule cost but transits the highest-risk corridors. Cost is roughly proportional to risk imposed — equitable.
- Small-vessel sector (35–65 ft) would bear new costs under PA 2022 without comparable risk-imposition data. Without per-class strike-risk evidence (NMFS RFI), this could be inequitable.
- Recreational sector currently externalizes ~\$0 of cost but contributes documented mortality near calving grounds. AIS-mandate equity question.

Benefit concentration:

- Public non-use benefits are diffuse (national/global willingness-to-pay).
 - Direct industry benefits accrue to whale-watching tourism (~\$10–30 M/year in NE/SE US) and to seafood industries via market-access certifications (MSC re-certification of Maine lobster, etc.).
-

8. Tail-Risk Adjustment (Why Headline B/C Understates the Speed Rule's Value)

The model assumes **zero probability of UME recurrence** in forward projections, which understates the speed rule's value. Adjusting for the indirect 35–45% probability of a UME-class regime within 15 years:

- **Expected loss without speed rule under climate stress:** roughly 25 additional whales / UME × 35% probability = 9 expected whales of incremental loss
- **Compounding over 50 yr (Poisson rate ~3%/yr):** expected ~15–25 additional whales
- **Adjusted speed-rule benefit:** +75 baseline + 15–25 climate-tail = **+90–100 whales over 50 year**
- **Adjusted B/C at \$3 M/whale:** (270–300 M benefit) / (\$1.5 B cost) $\approx$ 0.18–0.20 — still appears unfavorable on simple monetary terms

The case for the speed rule is fundamentally **risk-management, not cost-minimization**: the population cannot tolerate even one more UME within 15 years without crossing demographic thresholds where stochastic / Allee effects become dominant. Conventional B/C analysis underweights this tail risk.

9. Summary Recommendation

Most cost-effective package (sized to deliver $\geq 90\%$ of the speed rule's risk-management benefit at the lowest credible industry cost):

1. **Retain mandatory SMA speed rule** (≥ 65 ft baseline; extended to ≥ 35 ft per PA 2022) — \$30–46 M/yr industry cost
2. **Fund avoidance-maneuver field studies** (\$3 M/yr) — pays for itself in value-of-information
3. **Scale Whale Alert + expanded PAM/aerial coverage** (\$15 M/yr public) — B/C $2.3\times$
4. **Implement mandatory DSZ layer** when triggered by 1–3 (\$20 M/yr industry) — adjunct, not replacement
5. **Mandate AIS for 35–65 ft** (\$15 M/yr industry) — enables compliance verification of #4
6. **Incentivize Thermal/IR adoption** as complement (port-fee discounts $\approx$ revenue-neutral)

Estimated total annual industry cost: \$80–100 M/yr ($\approx 2\times$ current speed rule) for a **2–3 $\times$ increase in expected whale-population benefit** and significantly reduced tail risk.

Avoid: Any policy that treats Tier-2 onboard technology as a substitute for Tier-1 mandatory speed restrictions until avoidance-maneuver studies validate $P(\text{avoid}|\text{detect}) > 70\%$.

This portfolio is consistent with the unchanged `narw_model_v2a.py` calibration: it leverages the model's quantitative ranking of policy options (SL > Reg > Tech) and respects the 50% empirical-defensibility cap on avoidance-maneuver assumptions.